

# Business Summary Report: Predictive Insights for Collections Strategy

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## 1. Summary of Predictive Insights

Analysis of Geldium's dataset (500 customers, 19 variables) identified the following patterns and risk indicators to inform the Collections team's prioritization strategy.

### Top 3 Risk Factors:

- **Credit Utilization Rate:** Customers consistently using a high proportion of their available credit limit signal financial stress and reduced repayment capacity. In standard industry practice, utilization above 70% is considered a leading indicator of delinquency risk.
- **Missed Payment History:** Customers with 2 or more recorded missed payments show a compounding risk pattern — a single missed payment often precedes further default, particularly in the 30–90 day delinquency window Geldium is targeting.
- **Debt-to-Income (DTI) Ratio:** A high DTI indicates a customer's monthly obligations are disproportionate to their income, limiting their ability to absorb unexpected financial shocks and increasing delinquency likelihood.

Important note for the Collections team: Our EDA found that the above risk factors align with industry best practice; however, the current dataset's target variable (Delinquent\_Account) showed weaker-than-expected correlation with these features. We recommend that Geldium validates the labeling of the target variable against live repayment records before deploying model outputs operationally. This is flagged as a prerequisite in the modeling plan and does not diminish the value of the intervention strategy below.

Optional: Include a Key Insights Summary Table (you may create this in Excel or insert manually).

| Key Insight                                                           | Customer Segment                                                                     | Influencing Variables                                                                                      | Potential Impact                                                                                                                                       |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customers with 2+ missed payments in a 6-month window are at elevated | Customers aged 18–35 with short account tenure (<2 years) and limited credit history | Missed_Payments, Credit_Utilization, Debt_to_Income_Ratio, Account_Tenure, Month_1–Month_6 payment history | Prioritize proactive outreach to this segment before payments cross the 30-day threshold; offer structured payment plans or early intervention support |

|                                                   |  |  |  |
|---------------------------------------------------|--|--|--|
| risk of crossing the 30-day delinquency threshold |  |  |  |
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## 2. Recommendation Framework

Based on the insight that customers with 2+ missed payments and high credit utilization represent the highest-risk segment, the following SMART recommendation is proposed for Geldium's Collections team.

Use the following subheadings to guide your structure:

- Restated Insight: Customers with 2 or more missed payments in the past 6 months and a credit utilization rate above 70% are the segment most likely to cross the 30-day delinquency threshold and require Collections intervention.
- Proposed Recommendation: Launch a targeted early intervention outreach programme for high-risk customers, structured as follows:
  - Specific: Target customers flagged as high-risk by the predictive model (2+ missed payments AND credit utilization >70%) with a dedicated outreach sequence — beginning with an SMS reminder at Day 25 (5 days before the 30-day threshold), followed by a phone call at Day 28 for non-responders, offering a structured payment plan or short-term deferral option.
  - Measurable: Reduce 30-day delinquency rates among the targeted segment by 10–15% within the pilot period, measured by comparing the flagged group's delinquency rate against a matched control group not receiving the intervention.
  - Actionable: The Collections team implements the outreach using existing SMS and call-centre infrastructure. The AI model provides a daily updated prioritization list ranked by risk score, enabling agents to focus effort on the highest-risk cases first.
  - Relevant: Directly addresses Geldium's stated goal of improving repayment outcomes and reducing delinquency rates through proactive, data-driven customer engagement rather than reactive case handling.
  - Time-bound: 6-week pilot programme, with performance reviewed at Week 3 (interim) and Week 6 (final). Full rollout decision made within 2 weeks of pilot conclusion based on measured outcomes.
- Justification and Business Rationale: Early intervention at Day 25–28 targets the window where customer outreach is most likely to produce a positive response — before the customer has formally crossed into delinquency and before the financial and reputational cost of a missed payment is locked in. Focusing on a specific, model-defined segment (rather than broad outreach) reduces cost per contact and

avoids over-contacting low-risk customers, which can damage customer relationships. The pilot structure limits risk while generating the real-world performance data needed to refine the model's thresholds before full deployment.

### **3. Ethical and Responsible AI Considerations**

All AI-driven recommendations carry responsibilities around fairness, transparency, and customer impact. The following considerations apply to Geldium's implementation of this predictive model and outreach strategy.

- **Potential for bias or unfair treatment:** Two specific risks were identified in our analysis. First, Location and Employment\_Status may act as proxy variables for socioeconomic or demographic characteristics — meaning the model could disproportionately flag customers from certain areas or employment categories even when their actual repayment behavior does not differ materially. Mitigation: run a disparate impact analysis across these fields before deployment, and remove or adjust any variable found to introduce unfair segmentation. Second, younger customers (18–30) with short account tenure may be systematically over-represented in the high-risk segment simply due to limited credit history rather than genuine financial distress. Mitigation: apply age-group fairness checks and consider a separate risk calibration for thin-file customers rather than applying a single model threshold universally.
- **Explainability:** Logistic regression was selected precisely because its outputs can be translated into plain language for non-technical stakeholders. Each risk flag can be explained to the Collections team as: 'This customer was prioritized because they have missed 2 payments in the last 6 months and are using over 70% of their available credit — both of which are associated with a higher likelihood of missing their next payment.' No black-box outputs; every flag has a traceable, human-readable reason.
- **Responsible financial decision-making:** Model outputs should function as a decision-support tool, not an automated decision system. Every customer contacted through the outreach programme should have been reviewed by a Collections agent — the model prioritizes who to call, but a human makes the final call on what action to take and how to engage. Customers who feel they have been incorrectly flagged should have a clear and accessible route to raise a concern.
- **Additional ethical principles:** Customer data used in the model must be handled in line with applicable data protection regulations. The model's performance, fairness metrics, and feature weights should be reviewed quarterly and disclosed internally to risk and compliance teams. Any expansion of AI-driven decision-making beyond prioritization (e.g., automated payment plan offers) should be subject to a separate ethical review before implementation.

Keep your report under two pages. Write in plain, professional language. Use headings, bullet points, or short paragraphs to make it easy for the reader to follow.